

COURSE OBJECTIVES:

1. To understand the various components of full stack development
2. To learn Node.js features and applications
3. To develop applications with MongoDB
4. To understand the role of Angular and Express in web applications
5. To develop simple web applications with React

SYLLABUS

UNIT I BASICS OF FULL STACK 9

Understanding the Basic Web Development Framework - User - Browser – Webserver - Backend Services – MVC Architecture - Understanding the different stacks –The role of Express – Angular – Node – Mongo DB – React

UNIT II NODE JS 9

Basics of Node JS – Installation – Working with Node packages – Using Node package manager – Creating a simple Node.js application – Using Events – Listeners –Timers - Callbacks – Handling Data I/O – Implementing HTTP services in Node.js 95

UNIT III MONGO DB 9

Understanding NoSQL and MongoDB – Building MongoDB Environment – User accounts – Access control – Administering databases – Managing collections – Connecting to MongoDB from Node.js – simple applications

UNIT IV EXPRESS AND ANGULAR 9

Implementing Express in Node.js - Configuring routes - Using Request and Response objects - Angular - Typescript - Angular Components - Expressions - Data binding - Built-in directives

UNIT V REACT 9

MERN STACK – Basic React applications – React Components – React State – Express REST APIs - Modularization and Webpack - Routing with React Router – Server-side rendering



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LAB EXPERIMENTS:

1. Develop a portfolio website for yourself which gives details about yourself for a potential recruiter.
2. Create a web application to manage the TO-DO list of users, where users can login and manage their to-do items
3. Create a simple micro blogging application (like twitter) that allows people to post their content which can be viewed by people who follow them.
4. Create a food delivery website where users can order food from a particular restaurant listed in the website.
5. Develop a classifieds web application to buy and sell used products.
6. Develop a leave management system for an organization where users can apply different types of leaves such as casual leave and medical leave. They also can view the available number of days.
7. Develop a simple dashboard for project management where the statuses of various tasks are available. New tasks can be added and the status of existing tasks can be changed among Pending, In Progress or Completed.
8. Develop an online survey application where a collection of questions is available and users are asked to answer any random 5 questions.

COURSE OUTCOMES:

CO	COURSE OUTCOME	K LEVEL
At the end of the course student will be able to		
CO 1	Understand the various stacks available for web application development	K2
CO 2	Use Node.js for application development	K2
CO 3	Develop applications with MongoDB	K2
CO 4	Use the features of Angular and Express	K2
CO 5	Develop React applications	K2

CO's-PO's & PSO's MAPPING

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	3	2	0	0	0	0	0	0	1	1	1	1	1	1	0
CO 2	3	3	3	3	2	1	0	0	2	0	2	3	2	2	1
CO 3	3	3	2	2	2	2	0	0	2	2	2	3	2	2	1
CO 4	3	3	2	2	2	1	0	0	1	1	2	2	2	1	1
CO 5	3	3	3	3	3	1	0	0	0	0	2	2	2	2	1
Avg	3	3	2	2	2	1	0	0	1	1	2	2	2	2	1

3 Strongly Agreed

2 Moderately Agreed

1

Reasonably Agreed

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TEXT BOOKS

1. Brad Dayley, Brendan Dayley, Caleb Dayley, 'Node.js, MongoDB and Angular Web Development', Addison-Wesley, Second Edition, 2018
2. Vasan Subramanian, 'Pro MERN Stack, Full Stack Web App Development with Mongo, Express, React, and Node', Second Edition, Apress, 2019.

REFERENCES

1. Chris Northwood, 'The Full Stack Developer: Your Essential Guide to the Everyday Skills Expected of a Modern Full Stack Web Developer', Apress; 1st edition, 2018
2. Kirupa Chinnathambi, 'Learning React: A Hands-On Guide to Building Web Applications Using React and Redux', Addison-Wesley Professional, 2nd edition, 2018

E-LEARNING MATERIALS:

1. <https://www.coursera.org/specializations/full-stack-react>
2. <https://www.udemy.com/course/the-full-stack-web-development/>



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